

Sophia S. Chen

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SKILLS

- Languages & Tools: Python, Java, SQL, R, Visual Studio Code, Jupyter, Git/SVN, Docker
 - Frameworks: Numpy/Pandas, Matplotlib, FastAPI, OpenCV
 - CAD/Simulation: Autodesk Inventor, SolidWorks Drawings, Cura 3D Slicer, LTSpice, Matlab
 - Prototyping/Lab: 3D Printing, Tormach CNC, Metal/wood shop (lathe, router, bandsaw), scope, soldering
 - Fluency: English and Chinese
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EDUCATION

Rice University, Houston, TX

Graduation: 5/2029

- **Major:** Computer Science (Minor: Data Science), GPA 3.9/4.0
 - **Coursework:** Algorithmic Thinking, Data Science Tools & Models (PostgreSQL, NumPy/Pandas/SKLearn, Spark), Comp390 Projects (Computational Genomics), Comp Thinking (Python), Statistics for Data Science (R), Prompt Engineering, Economics, Multivariable Calculus, Linear Algebra, Physics
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WORK EXPERIENCE

Computer Engineering Intern, NASA Armstrong Flight Research Center, Edwards, CA

8/26 - 12/26

- Develop a search microservice in Java/Spring Boot with Elasticsearch to implement semantic (natural language) vector retrieval over engineering report archives.
- Generate synthesized summaries of top search results using an LLM-driven RAG pipeline with Spring AI.
- Plan and execute hardware-in-the-loop HIL/SIL testing at NASA sites within Edwards Air Force Base
- Support Range Operations, Range Engineering, and mission control room execution to ensure manned aircraft airworthiness and completion of in-flight experiments.

Research Assistant, Treangen Lab, Rice University, Houston, TX

5/26 - 7/26

Undergraduate Researcher (part-time), Treangen Lab, Rice University, Houston, TX

9/25 - 5/26

- Worked on a computational genomics workflow to classify microbiome samples using a variational autoencoder (Python, Scikit-Learn) trained on next-generation long-read sequencing data.
- Deployed and managed jobs on Rice's high-performance computing (HPC) clusters (RAPID and NOTS)
- Used AI agents to code/debug (Python) and validate novel LLM-based approach for genome variant sequencing (multiplex PCR tiling).

PROJECTS

- MediMetrics (HackRice 2025): Applied OpenCV and MediaPipe (ML) for medical landmark measurement for Electronic Health Records. Our hackathon submission included a full-stack FastAPI imaging backend (Ngrok localhost), React frontend on Vercel and Firebase authentication/storage.
- Electromagnetic Launcher (2023-24): Developed new coilgun design with Arduinos, gyro sensors, H-bridge and electromagnets to improve coilgun efficiency. Won 2024 regional and 4th in EE at state.

LEADERSHIP

HackRice Organizing Team, Rice University, Houston, TX

1/26 - now

- Organize workshops, welcome activities, and competition logistics for worldwide teams at Rice hackathons.

Lynbrook FIRST Robotics, San Jose, CA

9/21 - 5/25

- Vice President of Robot Design & Build (2024-5): Led robot design and integration, HW/SW/EE for 50+ team.
- Overall Design Lead (2023-24): Trained team in Autodesk tools and Lynbrook Design Methodology.
- Subsystem Lead - Drive train (2022-3), Shooter/Intake (2023-4), End Effector (2024-5): Design/CAD/CAM/test.

Activities: HackRice (organizer/participant), Rice Robotics Club, SWE@Rice